

AI Research Impact Observatory

Sponsorship: \$1,000 to winner | \$500 to runner-up

TL;DR

Machine learning is transforming science, but **how much, where, and with what consequences?**

Some claim AI is revolutionizing drug discovery and materials science. Others worry it's creating unreplicable results and citation bubbles. The truth is: **we don't actually know the quantitative impact.**

Your challenge: Build systems that measure ML's real influence on scientific progress across disciplines.

The Dataset

Two-Tier Data Infrastructure

Validation Dataset

Test your metrics against expected results with our scientific validation dataset spanning computer science and domain sciences. More information on the metadata structure below in Data Access. Some sample data in this format that has not been passed through evaluation is provided! More detail in Data Access below.

“Scaling” Dataset

Build and scale your actual metrics using real data from open access corpuses from sources like arxiv, PubMed, etc.

What's Included

- **Domain science papers across:**
 - Drug discovery
 - Materials science
 - Climate science
 - Physics

- Neuroscience
- More!

Dataset Access

Validation Set(s): Check the directory for `modded\_s2orc\_data\_validation.jsonl.zip`. This file contains the validation dataset in the form of a .jsonl line entries. You will also find (A) `modded\_s2orc\_ml.jsonl.zip`, which contain example jsonl entries that 1) have not been run through evaluation and 2) are based on AI-enabled methods. (B) `modded\_s2orc\_nonml.jsonl.zip` which contain example jsonl entries that 1) have not been run through evaluation and 2) are not based on AI-enabled methods. (B)

Scaling Data: To produce the scaling dataset, scrape any desired data from open access corpuses like arXiv, PubMed, OpenRxiv, or any other desired sources. Data for these sources can usually be found on Kaggle (<https://www.kaggle.com/datasets/Cornell-University/arxiv>), in public google buckets, or simply accessed using an API-key. The data you use here (assuming it is open access) is completely up to you!

Metadata Structure: See the file “evaluation_structure” for an example of the metadata structure for evaluations.

Insight Goal

Quantify machine learning's real impact on scientific progress across disciplines. Measure discovery acceleration, adoption patterns, reproducibility effects, and discipline-specific outcomes.

Key objectives:

- Distinguish transformational ML applications from hype
 - Identify which fields benefit most
 - Determine which ML techniques have staying power
 - Assess whether AI-enabled research is more or less reproducible than traditional methods
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Build Objective

Create an interactive research intelligence platform that includes **one or more** of these capabilities:

1. Quantify ML Impact

Measure how much ML actually contributes to scientific breakthroughs and discovery efficiency.

- **Attribution scoring:** What % of breakthrough comes from ML vs. domain insight?
- **Acceleration metrics:** Did ML speed discovery by months/years?
- **Efficiency measures:** Cost per discovery

2. Visualize Adoption Dynamics

Show how ML techniques spread across scientific disciplines over time.

- Interactive dashboards showing S-curves and citation flows from ML methods to domain papers
- Animated temporal evolution (2016-2024)
- Cross-discipline comparison views

3. Analyze Quality Trade-offs

Investigate whether ML adoption correlates with better or worse research reproducibility.

- Cross-reference ML adoption with reproducibility rates
- Retraction pattern detection and signals
- Code/data availability correlations

4. Trace Discovery Impact

Follow the path from ML method development to real-world scientific impact.

- High-profile case studies (AlphaFold, COVID drug discovery)
- Journey tracking: ML method → domain application → real-world impact

Evaluation Criteria

Submissions will be evaluated on:

- **Impact measurement accuracy:** How well do your metrics align with expected results in the validation dataset?
- **Insight depth:** Does your analysis reveal novel, actionable insights about ML's impact on science?

- **Visualization quality:** Are adoption dynamics and patterns clearly communicated?
 - **Technical rigor:** Is your methodology sound and reproducible?
 - **Scalability:** Can your approach handle the full 20M+ paper dataset?
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Getting Started

1. Explore the validation dataset to understand the data structure
 2. Develop and test your metrics on the validation set
 3. Scale your solution to the production dataset.
 4. Build your interactive platform
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Good luck! 🚀